

ELEC3106 Lab Project Report

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Introduction

The objective of this report is to illustrate our design of a negative supply voltage generator, which produces a steady output supply voltage between $-4V > V_{OUT} > -6V$ with a 5V power supply and ground. Our design makes use of a 555 IC timer, a +5V power supply, as well as several Schottky Diodes and transistors.

Measurements

1. Layout

The schematic for our circuit, with resistor and capacitor values, can be seen in the appendix below. The simulation results are also included following the schematic. It may be noted that there are some differences between our schematic and our real implementation; this is caused by simulation limitations as discussed in lab report 4. For example, the existence of an 'ideal' power supply in simulations rather than one displaying voltage drops in real life. In addition to the negative power supply generator, we have also included several extensions:

1. Implementing a clock generator;
2. Output as close to -5V as possible;
3. Minimizing output voltage ripple.

Attempts were also made at implementing an early-warning drop-out supply, the final design of which, although not working, shall also be included in this report.

We will not begin discussing several segments of our circuit.

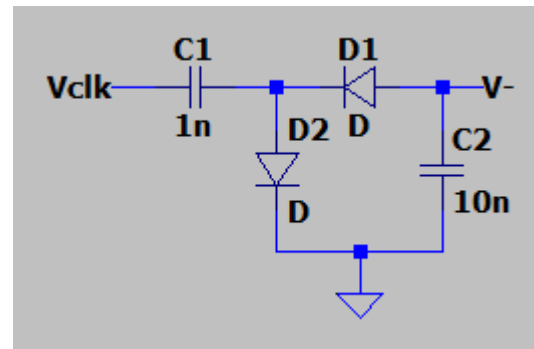


Fig 1.1. Our circuit's main body.

The main body of our circuit, as shown above, is responsible for generating the -5V power supply from the 555 IC clock timer. The clock output is to be connected to the port labelled 'Vclk', and the negative power supply can be taken from the port labelled 'V-'. When Vclk is positive, the capacitor is charged and the voltage across C1 will attempt to reach 5V, which is the clock's maximum voltage. However, the diode D2 will begin conducting when the voltage across C1 becomes 0.48V, so C1 will never be able to reach 5V across it. Therefore, one of C1's plates - the one closer to Vclk - is kept at 5V and the other plate is only ever able to reach 0.48V. After roughly 0.011ms, the clock input will revert to 0V from 5V. At this time, C1 will attempt to keep the potential difference of $5V - 0.48V = 4.52V$ across it, so the plate further away from Vclk will drop to -4.52V to maintain this coupling. D2 is unable to conduct this, so the negative voltage remains on C1.

At the same time, C2 is charging as well. When the voltage across C2 reaches 0.48V diode D1 will similarly conduct this away, leaving only 0.48V at the top plate of C2. So when the input Vclk goes to 0V, the voltage across C2 will drop to -4.02V in order to maintain the potential difference. This cycle will continue each time Vclk oscillates between 0V and 5V, thus producing our negative rail supply.

It could also be noted that charge pumps could be added to the end of the circuit in order to 'force' the voltage downward even more. Previous attempts using charge pumps created a rail supply of up (down) to -8V, which can then be manipulated to be closer to -5V using a voltage divider to ground.

The values of our capacitors for C1 and C2 were chosen to ensure that the time taken for the negative rail power supply to reach -5V (as shown in the simulation at the end of this report) will not be too long. Higher values of C1 increases the time spent to reach -5V, while lower values for C2 decreases stability for our negative rail when a load is attached.

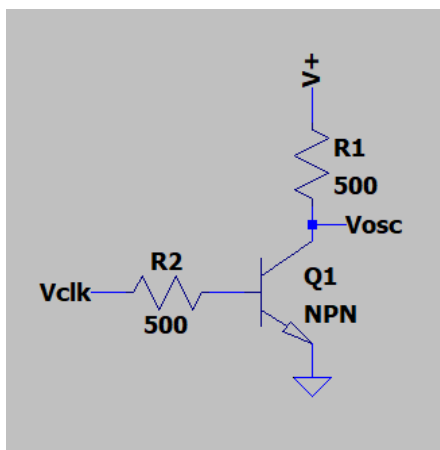


Fig 1.2. Incorporation of a BJT into a circuit.

A different part of our circuit that should be noted is the incorporation of BJT immediately after the 555 IC, and before the main body. As shown, the 555 IC output is connected to the BJT base via a 500Ω resistor, and the voltage at the collector (Vosc) is to be used as input to our main body. Unfortunately, this segment does not seem to be functional in our schematic - we assume that there is an issue with the transistor model. However, we can assure you that it does work in real life.

The function of this BJT is to ensure that there is enough current and/or voltage driving the circuit - especially when a load is attached. We included this BJT after seeing several of our previous circuit models have a 'drop' to -2V at the output negative rail supply V- when a load is attached. We theorize that the reason for the voltage 'drop' when a load is attached is a lack of current to drive the voltage; this was confirmed by the LM741 datasheet requiring supply currents I_{CC} to be 1.5mA or 2.8mA at most. When I_{EE} was not adequate the LM741 took voltage from V_{EE} to produce more current, using Ohm's Law $V=IR$. The BJT was included to prevent this, to ensure that there is enough current driving voltage throughout the circuit, even when a load is attached.

In the BJT segment, resistors of 500Ω were chosen so that current flow will not be overly restricted, while still ensuring that overcurrent does not occur.

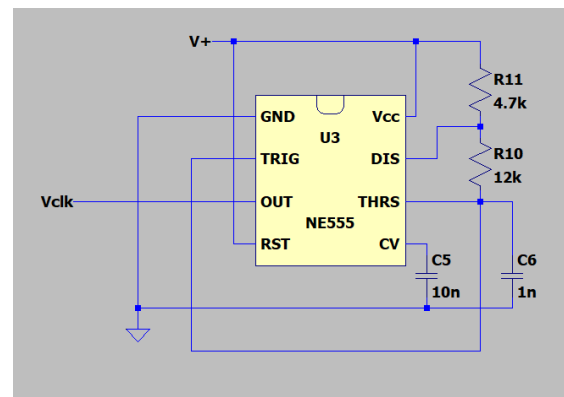


Fig 1.3. A 555 IC used to implement a clock generator.

The next segment to be discussed is the clock timer, which we implemented using the 555 IC. The values of R10, R11, and C6 were chosen to ensure that the clock output had the following characteristics:

- Roughly 50kHz frequency;
- A duty cycle of 58.19%;
- A high time of 0.001159ms.

These values were calculated using the formulas $f = \frac{1.44}{(R_1 + 2R_2)C}$, high time $T = 0.694(R_1 + R_2)C$ and low time $T = 0.694R_2C$ that we found.

The last part of our circuit to be discussed is the failed early drop-out warning. This circuit was to function using the LM339 comparator. The inverting input of the comparator was to be connected to the reference voltage.

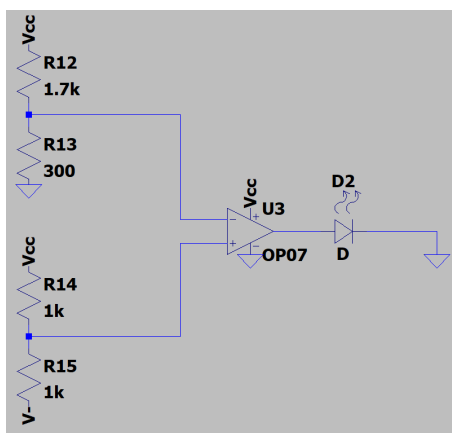


Fig 1.4. Early warning drop-out circuit.

The principle of this circuit was that instead of directly giving a warning when the negative rail supply surpassed a certain value, it would instead monitor the changes in V_- based on a reference voltage. The reference voltage we used was a 'change' of 1.5V - this voltage was obtained using a voltage divider from V_{cc} to GND. When V_- increased by more than 1.5V, a warning would be outputted. This should work due to the equal resistances of the resistors in the voltage divider between V_{cc} and V_- . When their values are +5V and -5V respectively, the non-inverting input will be 0V (smaller than the inverting input of 0.75V) and the comparator output is off; the LED will not light up. But when V_- increases by 1.5V or more, the non-inverting input will increase by 0.75V, so the inverting input will no longer be larger than the non-inverting

input. The comparator output will allow the LED to light up. It can also be speculated that the comparator output could probably be used with the reset pin (pin 4) on the 555 IC to completely halt the circuit.

We were unable to get this part working, but the resistor values R14 and R15 were chosen to be equal for easier calculation. This way, the non-inverting input would be exactly in the middle of V_{cc} and V_- . The values R12 and R13 were chosen so that the inverting input, which is our reference voltage, would have a value of 0.75V. This way, when V_- rises from roughly -5V to -5V + 1.5V = -3.5V, the voltage at the non-inverting input will change as well, from 0V to 0.75V.

2. Supply characterisation

We tested the negative supply generator under zero, 1k Ω , and 10k Ω load conditions. The output voltage range is $-4V > V_{out} > -6V$. With no load connected to the supply generator, the $V_{oc} = -5.56V$.

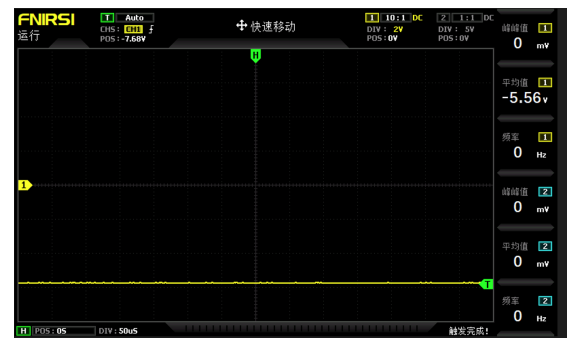


Fig. 2.1. Output voltage with no load

Then we measured the Voltage ripples, output voltages and output current for 1k Ω and 10k Ω loads.



Fig. 2.2. Output voltage with 1kΩ load

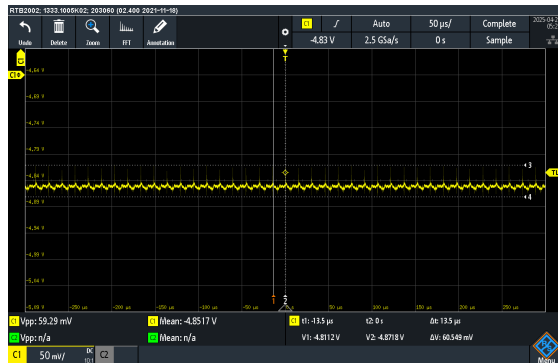


Fig. 2.3. Output voltage with 10kΩ load

Load	V_o	I_o	V_{ripple}
1kΩ	-4.1V	-23.5mA	104.9mV
10kΩ	-4.85V	-20.3mA	60.76mV
Op-Amp	5.884V	17.2mA	123mV

When supply generator is connected to the 1kΩ load, the output voltage was -4.06V with a voltage ripple of 146.51mV. The output current is 23.5mA; hence, we can calculate the output resistance using this equation:

$$R = \frac{V_{oc} - V_{Load}}{I} = \frac{(5.56V - 4.06V)}{23.5mA} = 63.829\Omega$$

When connected to the 10kΩ load, the output voltage was -4.85V with a voltage ripple of 60.76mV. The output current is 20.3mA; thus and the output resistance is

$$R = \frac{5.56V - 4.06V}{20.3mA} = 73.89\Omega$$

Notably, the output resistance differed for each load. This likely occurs because the capacitor's charge-discharge behaviour

changes; with the 10k load, the capacitor fully discharged each cycle, which yields a larger apparent R_{out} .

The power efficiency is calculated using this equation:

$$eff = \frac{V_o * I_o}{V_{in} * I_{in}}$$

Load	V_{in}	I_{in}	P_{eff}
1kΩ	5.4V	31mA	57.6%
10kΩ	5.4V	27mA	67.5%
Op-Amp	5.4V	27mA	69.4%

3. Functional test

The designed circuit is tested on its operation using LM741 with a 10kΩ output load resistor to ground. As can be seen in Figure 3.1, channel one in yellow shows the output of our designed circuit, V_o , supplied to the op-amp. It is measured to be around -4.65V. And channel two in green illustrates the output of the non-inverting op-amp. The noise at the op-amp output can be observed in Figure 3.2, as there are small amounts of fluctuations at the output signal (channel 2, blue).

In Figure 3.1, there are periodic sawtooth waves and spikes observed at the supply. The noise caused by the supply to the op-amp is measured to be around 124.46mV and the period of the sawtooth waves is measured to be about 95.5μs while the period of the spikes is approximately 13.64μs.

In addition, the supply is experiencing small-signal transients caused by large-signal transients as shown in Figure 3.1. The periodic peaks of the supply are aligned with the peaks of the op-amp

output, which is the transients introduced to the supply whenever the op-amp output swings rapidly. This might be due to the inverting charge pump's internal switching. As the capacitors cause voltage drops during charging and discharging, which leads to transient spikes and might pick up coupled noise. This noise should be able to be smoothed out in a number of ways; for example, filters can be used to remove noise.

switch-mode design may improve it further more. These measurements were carefully taken; but some error may arise during the demonstration or measurement period due to human error. This applies to our theory as well - we have tried our best.

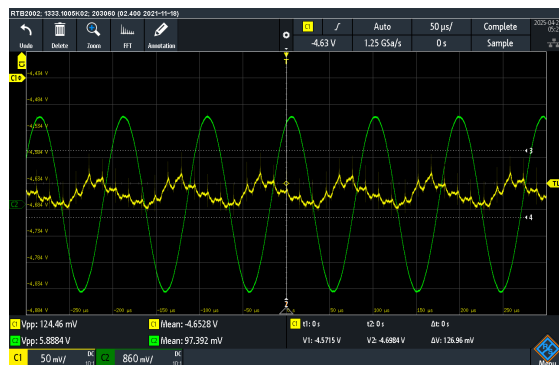


Figure 3.1: Vout supplied to the op-amplifier and output of the op-amp (with sine wave input signal)

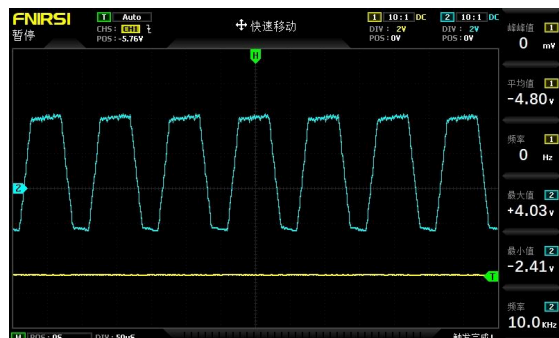


Figure 3.2: Vout supplied to the op-amplifier and output of the op-amp (with square wave input signal)

Conclusion

It can be seen that our negative power supply generator is stable, exhibiting minimal voltage ripple. Besides, our highest efficiency is around 69.4% for op-amp. This design mainly relies on the diodes and capacitors generating the negative voltage, both incur losses. Therefore, we used schottky diodes to reduce the loss. An inductor-based

Appendix

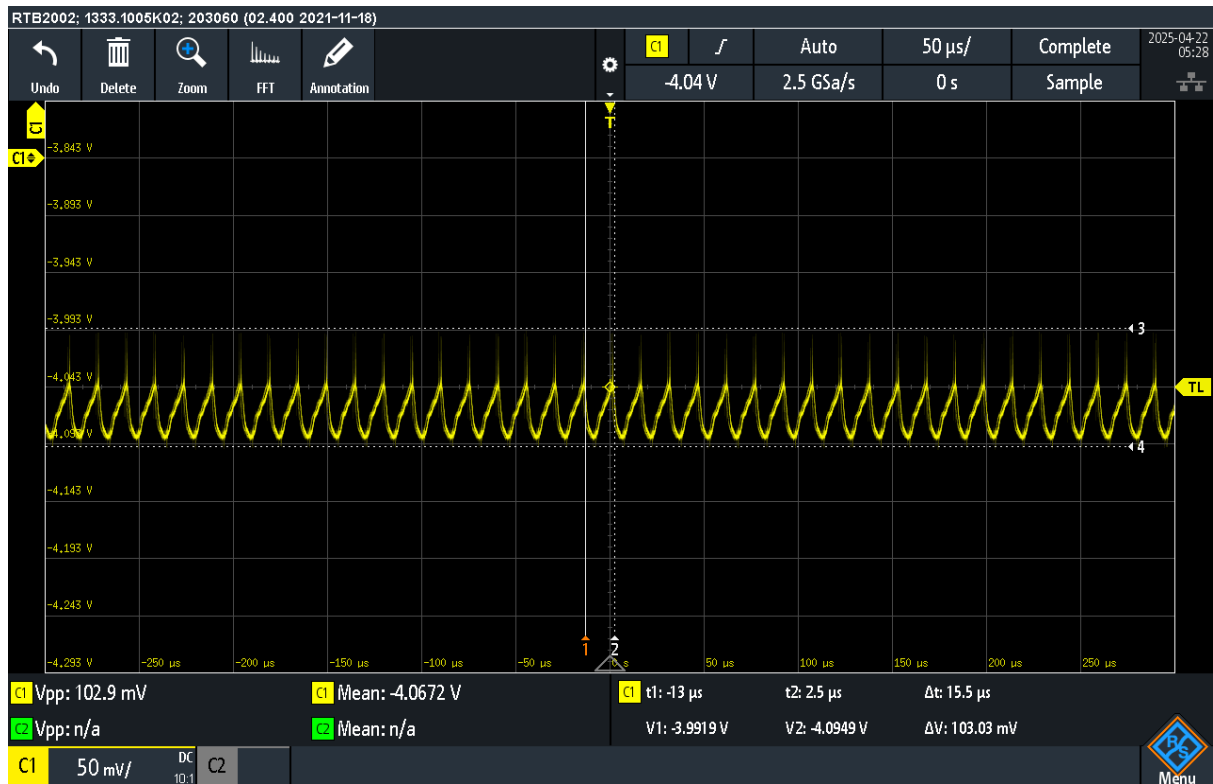


Figure: Voltage drop and V_o with the 1k load

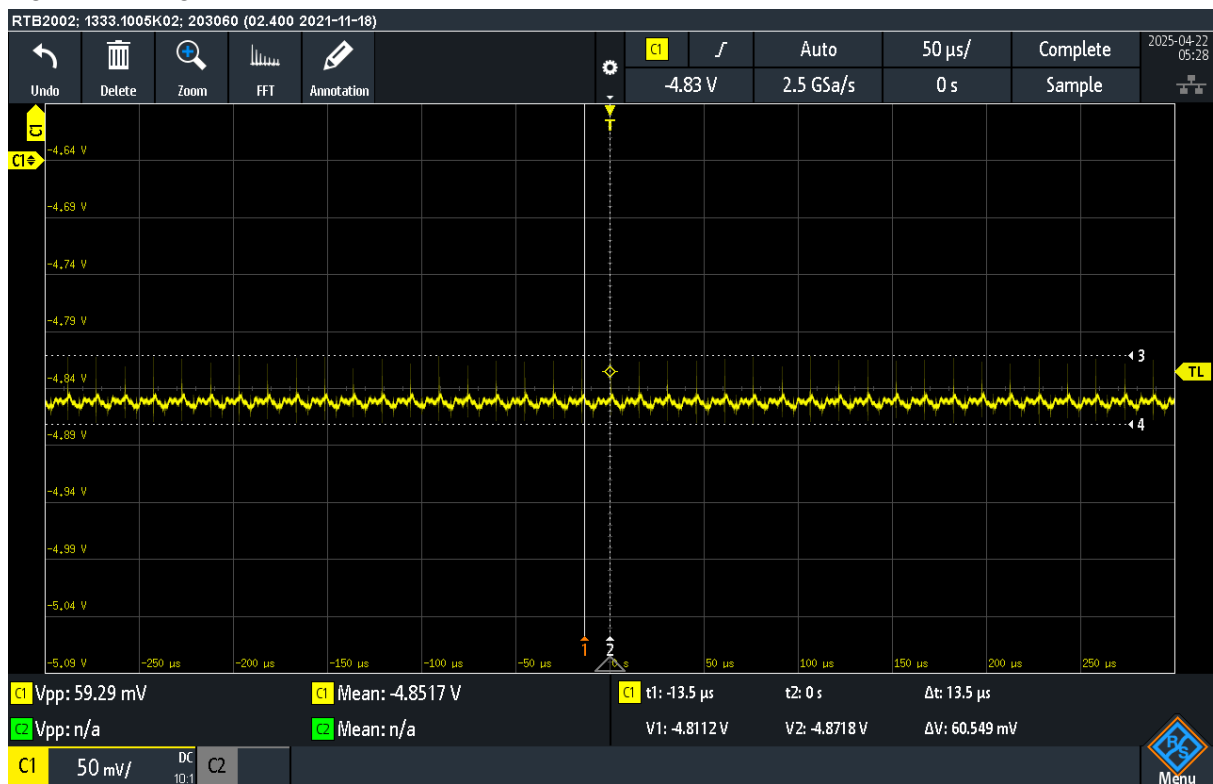


Figure: Voltage drop and Vo with the 10k load

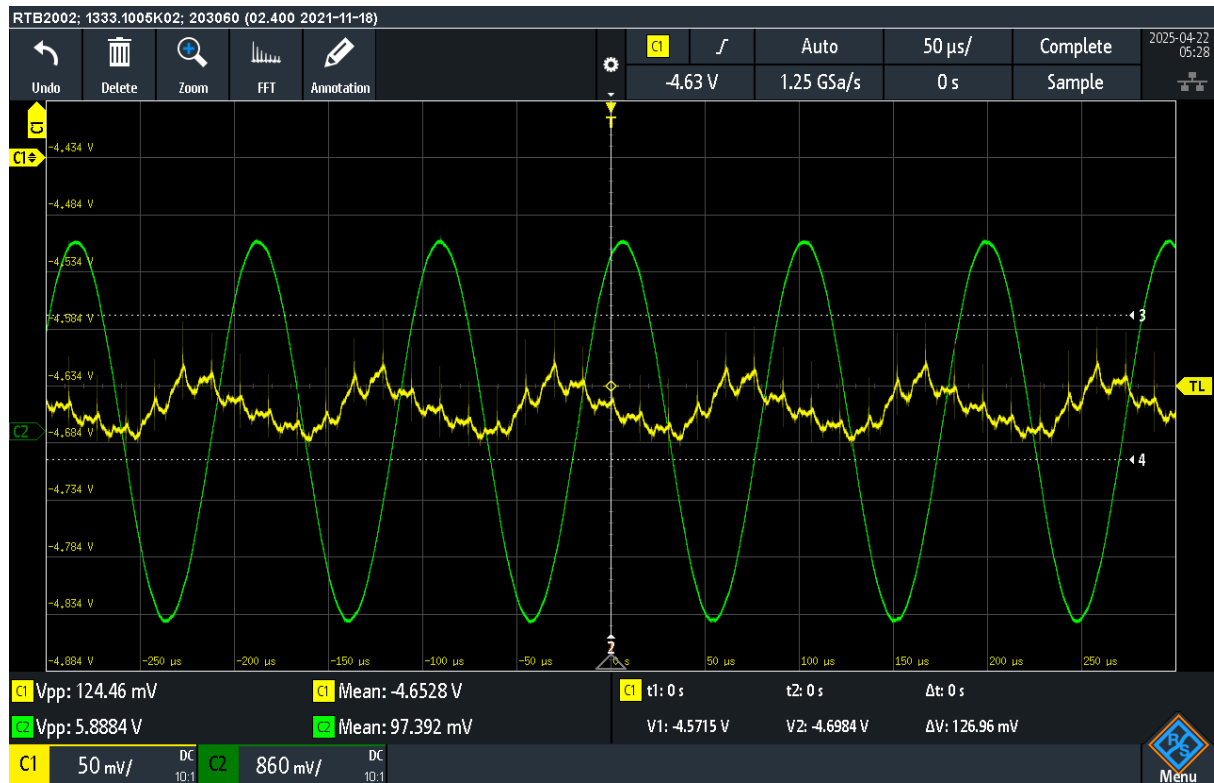


Figure: Voltage drop and Vo with the op-amplifier

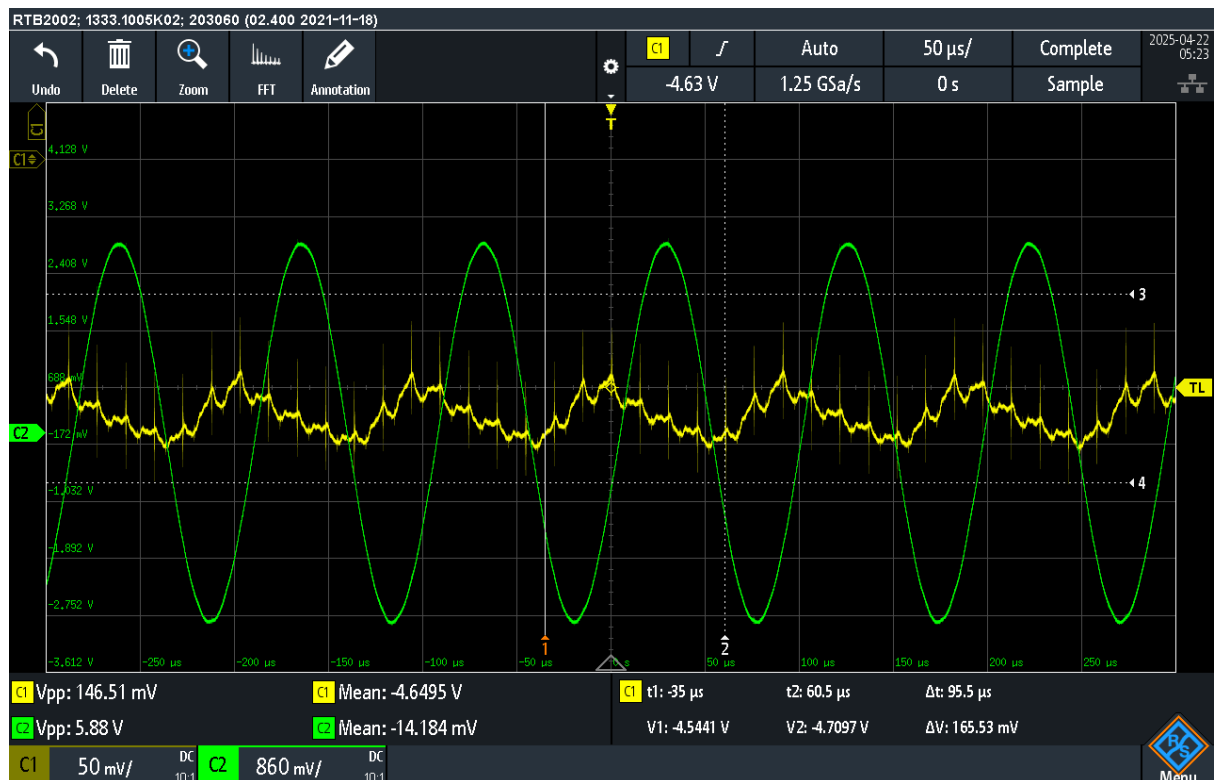


Figure: Voltage drop and Vo with the op-amplifier

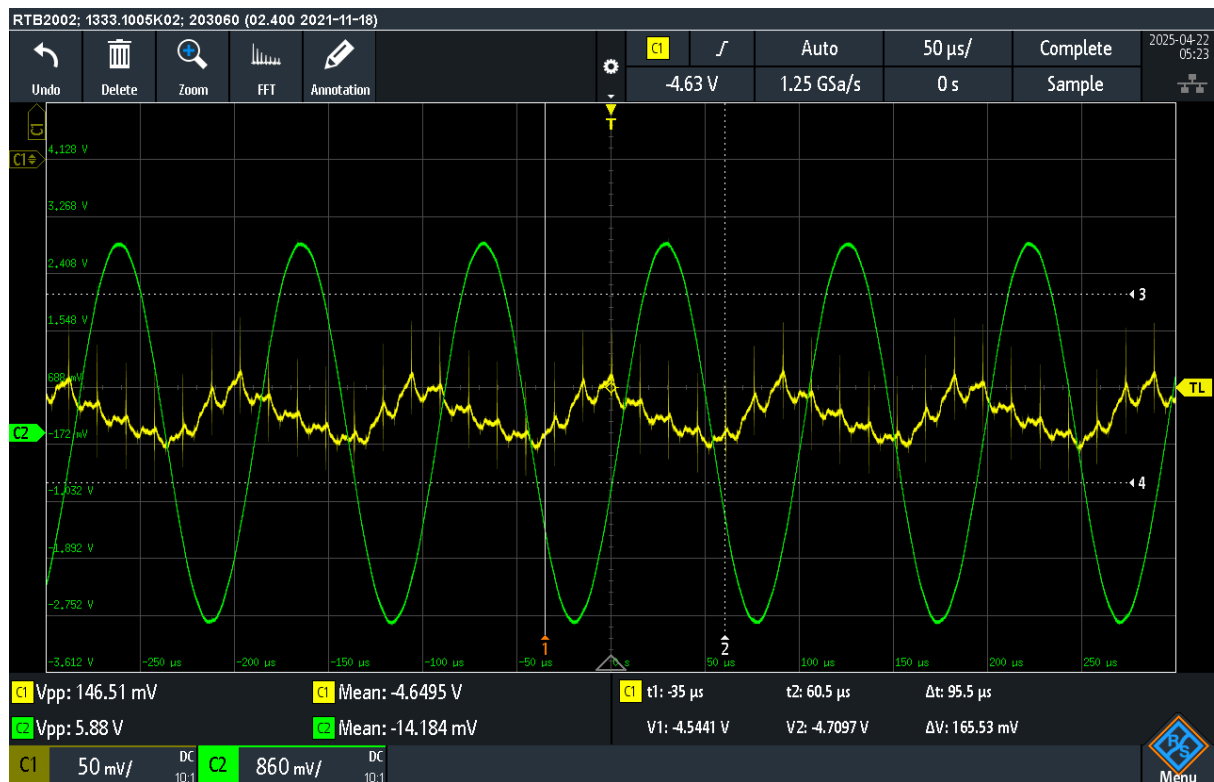


Figure: Voltage drop and V_o with the op-amplifier

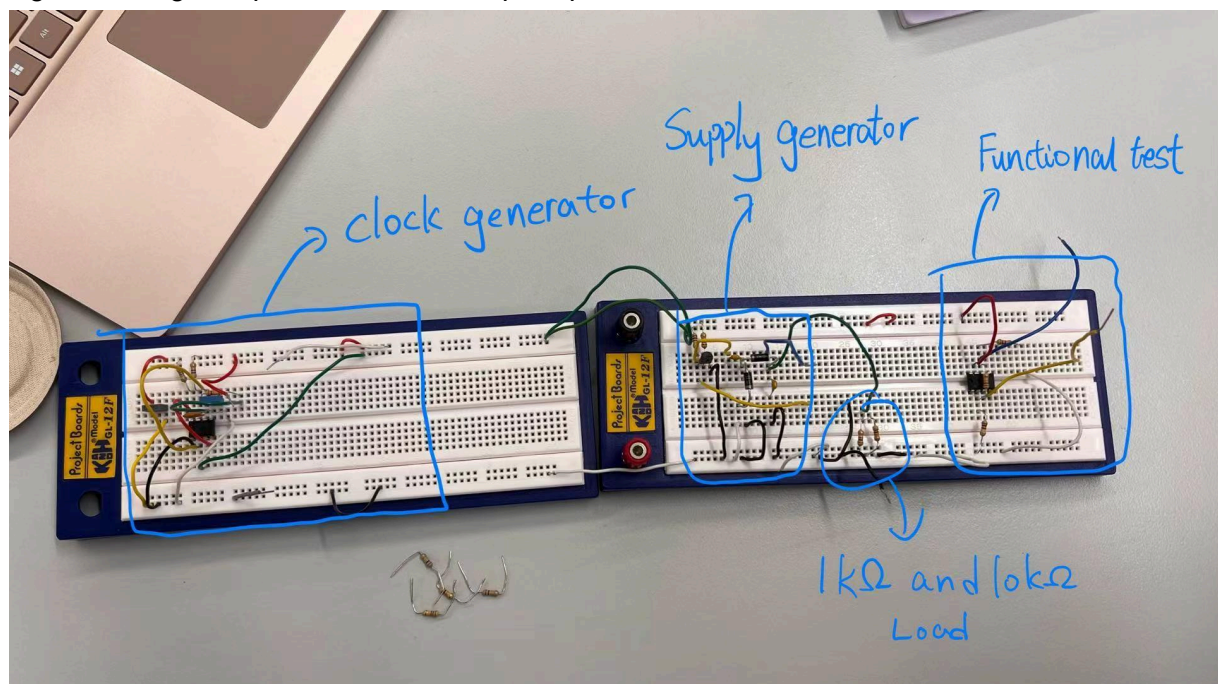


Figure: The layout of the physical circuit

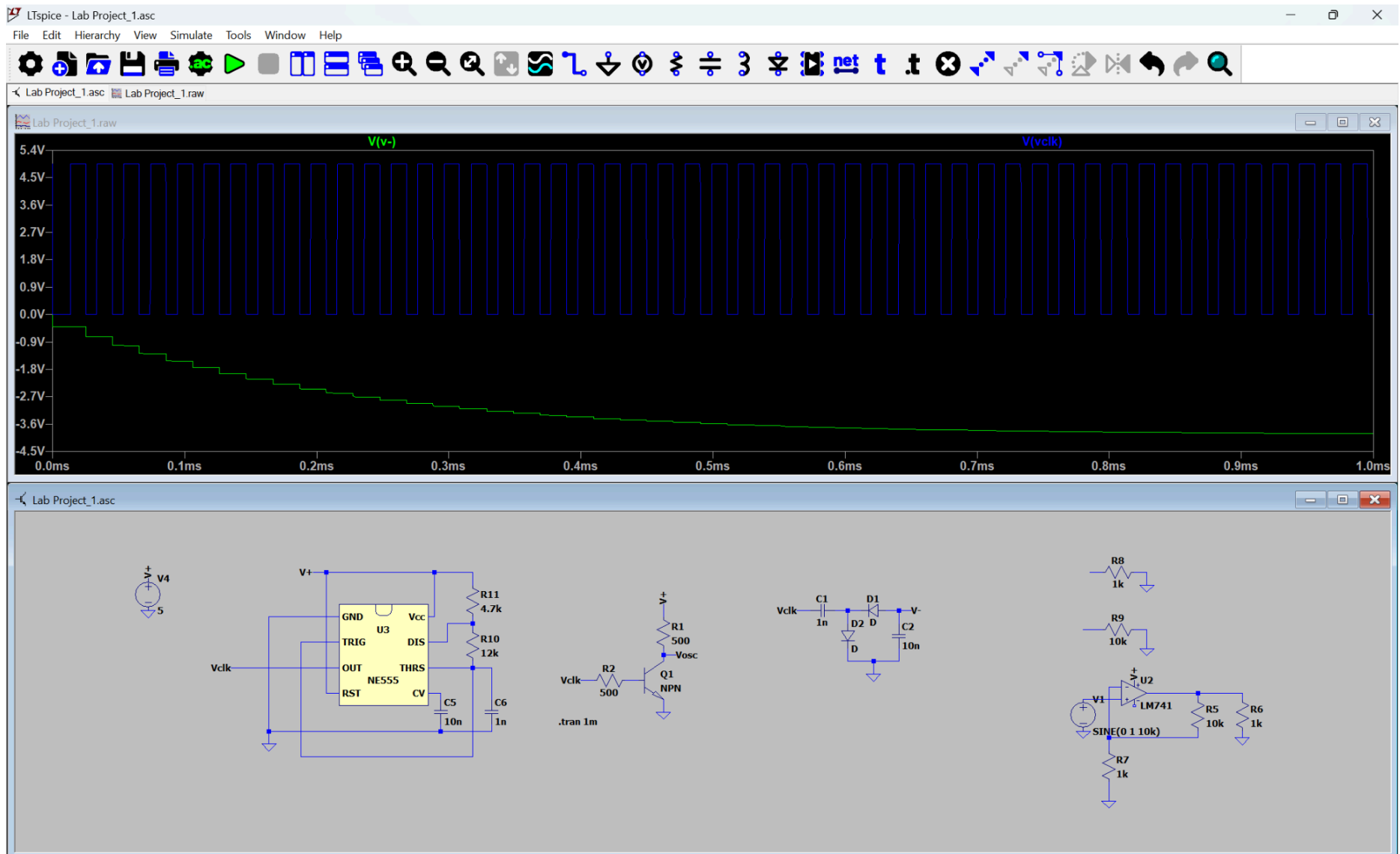


Figure: A schematic of our circuit. Some aspects may be changed in the real world implementation due to simulation limitations.